FREEDOM INTERNATIONAL SCHOOL

WORKSHEET - MCQ

PHYSICS

CLASS XII

ELECTROMAGNETIC WAVES

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1. A parallel plate capacitor is charged by a current of 2 x 10^{-7} A displaced between the plates of capacitor. When discharge of the capacitor takes place through a resistance, the rate of change of electric flux in (V m/s) will be
   (a) 2.26 x 10^4 (b) 4.26 x 10^8 (c) 3.26 x 10^6 (d) 6.26 x 10^9
2. A plane electromagnetic wave travels in vacuum along $\hat{k}$ direction, where $\hat{\imath}$, $\hat{\jmath}$ and $\hat{k}$ are unit vectors along x, y and z directions. The directions along which the electric and the magnetic field vectors point may be respectively
   (a) $\hat{\imath}$ and $\hat{\jmath}$ (b) $\hat{\imath}$ and - $\hat{\jmath}$ (c) $\hat{\jmath}$ and $\hat{\imath}$ (d) $\hat{k}$ and $\hat{\imath}$
3. Electromagnetic waves travel in a medium which has relative permeability 1.3 and relative permittivity 2.14. Then the speed of the electromagnetic wave in the medium will be
   (a) 13.6 x 10^6 m/s (b) 1.8 x 10^2 m/s (c) 3.6 x 10^8 m/s (d) 1.8 x 10^8 m/s
4. A plane electromagnetic wave of frequency 30 MHz travels in free space along the x- direction. The electric field component of the wave at a particular point of space and time is E= 6 V/m along y- direction. Its magnetic field component B at this point would be
   (a) 2 x 10^{-8} T along z- direction (b) 6 x 10^{-8} T along x- direction
   (c) 2 x 10^{-8} T along y- direction (d) 6 x 10^{-8} T along z- direction
5. Electromagnetic waves transport
   (a) charge and momentum (b) frequency and wavelength
   (c) energy and momentum (d) wavelength and energy
6. There are three wavelengths: 10^{-8} m, 10^{-2} m, 10^8 m. Their respective names are
   (a) visible rays, γ- rays, ultraviolet rays (b) ultraviolet, microwaves, radio waves
   (c) x- rays, visible rays, radio waves (d) radio waves, x- rays, micro waves
7. Human body radiates
   (a) micro waves (b) x -rays (c) infrared waves (d) gamma rays
8. The electric field associated with an electromagnetic wave in vacuum is given by $\vec{E} = 40\cos(kz - 6 \times 10^8 t)\,\hat{\imath}$, where E, z and t are in V/m, metre and seconds respectively. The value of wave vector k is
   (a) 2 m^{-1} (b) 0.5 m^{-1} (c) 6 m^{-1} (d) 3 m^{-1}
9. Which of the following statements is false for the properties of electromagnetic waves?
   (a) Both electric and magnetic field vectors attain the maxima and minima at the same place and same time
   (b) The energy in electromagnetic wave is divided equally between electric and magnetic vectors
   (c) Both electric and magnetic field vectors are parallel to each other and perpendicular to the direction of propagation of the wave

(d) These waves do not require any material medium for propagation

10. If λ_v, λ_x and λ_m represent the wavelength of visible light, x-rays and micro waves respectively, then
(a) $\lambda_m > \lambda_x > \lambda_v$ (b) $\lambda_m > \lambda_v > \lambda_x$ (c) $\lambda_v > \lambda_x > \lambda_m$ (d) $\lambda_v > \lambda_m > \lambda_x$

11. An electromagnetic wave of frequency 3 MHz passes from vacuum into a dielectric medium with permittivity $\varepsilon = 4$. Then,
(a) wavelength and frequency both remain unchanged
(b) wavelength is doubled and the frequency remains unchanged
(c) wavelength is doubled and the frequency becomes half
(d) wavelength is halved and the frequency remains unchanged

12. The ratio of amplitude of magnetic field to the amplitude electric field for an electromagnetic wave propagating in vacuum is equal to
(a) the speed of light in vacuum
(b) reciprocal of speed of light in vacuum
(c) unity
(d) the ratio of magnetic permeability to the electric susceptibility of vacuum

13. The electric field part of an electromagnetic wave in a medium is represented by $E_x=0$; $E_y= 2.5$ N/C $\cos[(2\pi \times 10^6 \text{ rad/s})\, t - (\pi \times 10^{-2} \text{ rad/m})\, x]$; $E_z=0$. The wave is
(a) moving along x- direction with frequency 10^6 Hz and wavelength and wavelength 100 m
(b) moving along x- direction with frequency 10^6 Hz and wavelength 200 m
(c) moving along y- direction with frequency 10^6 Hz and wavelength 200 m
(d) moving along y- direction with frequency $2\pi \times 10^6$ Hz and wavelength 200 m

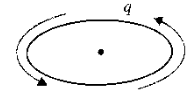
14. A parallel plate capacitor of capacitance 20 μF is being charged by a voltage source whose potential is changing at the rate of 3 V/s. The conduction current through the connecting wires, and the displacement current through the plates of the capacitor, would be, respectively
(a) 60 μA, 60 μA (b) 60 μA, zero (c) zero, zero (d) zero, 60 μA

For questions 15 to 20, two statements are given- one labelled Assertion (A) and other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.
D. Both Assertion and Reason are false.

15. **Assertion:** Electromagnetic radiations exert pressure.
Reason: Electromagnetic waves carry both momentum and energy.

16. **Assertion:** A charge moving in a circular orbit can produce electromagnetic wave.
Reason: The source of electromagnetic wave should be in accelerated motion.



17. **Assertion:** Microwaves are considered suitable for radar system.
Reason: Microwaves are of shorter wavelength.

18. **Assertion:** The ozone layer present at the top of stratosphere is very crucial for human survival.
Reason: Ozone layer prevents IR radiation.

19. **Assertion:** Gamma rays are electromagnetic waves having the smallest wavelength.
Reason: Gamma rays are having the lowest frequency.

20. **Assertion:** Ultraviolet radiations are used for LASIK eye surgery.
Reason: Being of shorter wavelength, UV radiations can be focused into very narrow beams for high precision applications.